

Grade 4 Mathematics

Մաթեմատիկա — 4-րդ դասարան

COURSE CODE	MTH-G4
PROGRAM	Elementary Mathematics · Տարրական մաթեմատիկա
LEVEL	Grade 4 · typical age 9–10
DURATION	72–84 instructional hours (approx. 66 lessons)
PREREQUISITE	Grade 3 Mathematics or equivalent: multiplication facts to 10×10 , fractions on a number line, area.
LEADS TO	Grade 5 Mathematics (MTH-G5)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 4 scales everything up. Place value extends to a million and, crucially, students learn that each place is ten times the one to its right — the insight that makes decimals possible. Multi-digit multiplication and division arrive, fraction arithmetic begins with addition and subtraction of like denominators and multiplication by a whole number, and decimals appear as a new notation for tenths and hundredths. Geometry becomes measured, with angles, and classification becomes precise.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Grade 4 is where procedural load rises sharply, and where students who memorized rather than understood in Grade 3 begin to fail. Վարժարան teachers therefore diagnose before they instruct: the placement assessment weights multiplication fluency and fraction meaning heavily, and gaps found there are closed before new content is layered on. Multi-digit algorithms are always introduced through the area model and partial products first, so that the compact algorithm is a shortening of something understood rather than a rule to obey.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Recalls multiplication facts to 10×10 in under 4 seconds each	Non-negotiable. Multi-digit multiplication with weak facts becomes an exercise in error accumulation.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Places fractions on a number line and explains equivalence with a model	If absent, run the Grade 3 Fractions Intensive concurrently with Units 1-2.
Adds and subtracts within 1,000 fluently, including across zeros	Prerequisite for the standard algorithm at larger magnitudes.
Finds the area of a rectangle and explains why the formula works	The area model is the vehicle for all multi-digit multiplication this year.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

- 1 Recognize that a digit in one place represents ten times what it represents in the place to its right.
- 2 Read, write, compare, and round multi-digit whole numbers to one million.
- 3 Fluently add and subtract multi-digit whole numbers using the standard algorithm.
- 4 Multiply a four-digit number by a one-digit number, and two two-digit numbers, using place-value strategies.
- 5 Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors.
- 6 Interpret a multiplication equation as a comparison and solve multiplicative comparison problems.
- 7 Find factor pairs, identify prime and composite numbers, and generate and analyze patterns.
- 8 Explain fraction equivalence with visual models and compare fractions with unlike denominators.
- 9 Add and subtract fractions and mixed numbers with like denominators; multiply a fraction by a whole number.
- 10 Understand decimal notation for tenths and hundredths, and compare decimals.
- 11 Measure and draw angles, and solve addition and subtraction problems involving angle measure.
- 12 Classify two-dimensional figures by properties of lines and angles, and identify lines of symmetry.

COURSE UNITS

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Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 • 6-7 LESSONS

Place Value to One Million

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TOPICS COVERED

- The ten-times relationship between adjacent places
- Reading and writing numbers to 1,000,000
- Standard, word, and expanded form for large numbers
- Comparing multi-digit numbers
- Rounding to any place
- Fluent multi-digit addition and subtraction
- Estimating sums and differences of large numbers

LEARNING OBJECTIVES

- 1 Explain that the 7 in 7,000 is ten times the 7 in 700.
- 2 Round a six-digit number to any specified place.
- 3 Add and subtract multi-digit numbers fluently using the standard algorithm.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- States the ten-times relationship correctly for any pair of adjacent places.
- Rounds correctly to a specified place in 9 of 10 trials.
- Computes multi-digit sums and differences at 90% accuracy.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Rounding is applied by looking at all following digits rather than the single next digit.	Use a number line and midpoint reasoning. The digit rule should be a summary of the reasoning, not a substitute for it.
Periods (thousands, millions) are not used when reading large numbers.	Teach reading in three-digit chunks with the period name attached; practice aloud.

TEACHING NOTE

The ten-times relationship is stated here for the first time as a general principle. Say it explicitly — it is what makes decimals in Unit 6 an extension rather than a new topic.

UNIT 2 • 9-10 LESSONS

Multi-Digit Multiplication

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TOPICS COVERED

- Multiplicative comparison: 'three times as many'
- Distinguishing additive from multiplicative comparison
- Multiplying by multiples of 10, 100, and 1,000
- The area model for two-digit by one-digit
- Partial products written vertically
- Four-digit by one-digit multiplication
- Two-digit by two-digit via the area model
- The standard algorithm as compressed partial products
- Estimating products to check reasonableness
- Multi-step word problems involving multiplication

LEARNING OBJECTIVES

- 1 Represent a two-digit by two-digit product with an area model and connect each region to a partial product.
- 2 Multiply four-digit by one-digit and two-digit by two-digit numbers accurately.
- 3 Distinguish 'five more than' from 'five times as many' in a word problem.
- 4 Estimate a product before computing and use it to check.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Draws a correct area model for a two-digit by two-digit product in 4 of 5 trials.
- Computes two-digit by two-digit products at 85% accuracy.
- Correctly classifies comparison problems as additive or multiplicative in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
In two-digit by two-digit multiplication only two of the four partial products are computed.	The classic error. The area model makes the missing regions visible — return to it whenever this appears, rather than repeating the algorithm.
The placeholder zero in the second partial product is omitted.	Do not teach 'add a zero'. Teach that the second row multiplies by 30, not 3, so the product is ten times larger.

TEACHING NOTE

Do not rush to the standard algorithm. Students who spend three lessons on the area model make dramatically fewer errors for the following two years.

UNIT 3 • 8-9 LESSONS

Division, Factors, and Patterns

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TOPICS COVERED

- Division with remainders and interpreting the remainder in context
- Dividing multiples of 10 and 100
- Partial quotients as a written strategy
- Four-digit dividends with one-digit divisors
- Checking division with multiplication
- Finding all factor pairs of a number to 100
- Multiples, primes, and composites
- Divisibility by 2, 5, and 10
- Generating a number or shape pattern from a rule
- Identifying features of a pattern not stated in the rule

LEARNING OBJECTIVES

- 1 Divide a four-digit number by a one-digit number using partial quotients or the standard algorithm.
- 2 Interpret a remainder correctly in context (drop it, round up, or report it as a fraction).
- 3 List all factor pairs of any number to 100 and classify it as prime or composite.
- 4 Generate a pattern from a rule and describe a property the rule did not state.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Divides accurately with remainders at 85% accuracy.
- Interprets remainders correctly in context in 4 of 5 word problems.
- Lists all factor pairs of a given number correctly in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The remainder is always discarded regardless of the question asked.	Use three word problems with identical numbers but different questions — how many buses, how many left over, what fraction remains — and let the contrast do the teaching.
Factors and multiples are confused with one another.	Anchor with language: factors are inside the number, multiples are beyond it. Use a factor rainbow and a multiple number line side by side.

TEACHING NOTE

Partial quotients is the more transparent method and should be secure before the standard algorithm is offered. Many students are best served by keeping partial quotients as their primary method through Grade 5.

UNIT 4 • 7-8 LESSONS

Fraction Equivalence and Comparison

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TOPICS COVERED

- Why multiplying numerator and denominator by the same number preserves value
- Generating equivalent fractions with models and with reasoning
- Simplifying fractions to lowest terms
- Comparing fractions with unlike denominators using benchmarks
- Comparing by creating common denominators
- Comparing by creating common numerators
- Ordering a set of fractions
- Fractions and decimals on the same number line

LEARNING OBJECTIVES

- 1 Explain why $\frac{3}{4} = \frac{6}{8}$ in terms of partitioning each part into two.
- 2 Compare two fractions with unlike denominators by at least two different methods.
- 3 Order a set of four fractions and justify the ordering.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Generates equivalent fractions correctly in 9 of 10 trials.
- Compares unlike-denominator fractions correctly in 9 of 10 trials with stated reasoning.
- Explains the equivalence rule with a model rather than by citing it.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child adds the same number to numerator and denominator to make an equivalent fraction.	Show with a model that $\frac{1}{2}$ and $\frac{2}{3}$ are not equal. Equivalence comes from re-partitioning, which is multiplicative, not additive.
Fractions are compared by numerator alone or denominator alone.	Require the student to state which method they are using and why it applies to this pair.

TEACHING NOTE

The equivalence rule must be justified, not announced. A student who can only recite 'multiply top and bottom' will apply it to addition next year.

UNIT 5 • 8-9 LESSONS

Fraction Operations

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TOPICS COVERED

- Decomposing a fraction into a sum of unit fractions
- Adding and subtracting fractions with like denominators
- Mixed numbers and improper fractions; converting between them
- Adding and subtracting mixed numbers with like denominators, including regrouping
- Multiplying a fraction by a whole number as repeated addition
- Word problems involving fraction addition, subtraction, and multiplication by a whole number
- Estimating fraction sums using benchmarks

LEARNING OBJECTIVES

- 1 Decompose $\frac{7}{8}$ into unit fractions in more than one way and record it as an equation.
- 2 Add and subtract mixed numbers with like denominators, regrouping when necessary.
- 3 Multiply a fraction by a whole number and interpret the product in context.
- 4 Estimate whether a fraction sum is closer to 0, $\frac{1}{2}$, or 1 before computing.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Adds and subtracts like-denominator fractions and mixed numbers at 90% accuracy.
- Converts between mixed and improper forms in 9 of 10 trials.
- Solves fraction word problems correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Denominators are added along with numerators: $\frac{1}{4} + \frac{1}{4} = \frac{2}{8}$.	Return to the unit-fraction meaning: one fourth plus one fourth is two fourths, because the <i>size</i> of the piece did not change. The number line makes this unmistakable.
When subtracting mixed numbers, regrouping is done as if the fraction part were base ten.	Regroup one whole into the correct number of fractional parts — $\frac{3}{5}$ becomes $2\frac{6}{5}$, not $2\frac{11}{5}$. Model with fraction strips.

TEACHING NOTE

Only like denominators this year. Resist parental pressure to introduce unlike denominators early; Grade 5 does it properly, and premature exposure produces the add-the-denominators error.

UNIT 6 • 8-9 LESSONS

Decimals, Measurement, and Geometry

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TOPICS COVERED

- Tenths and hundredths as fractions and as decimals
- Decimal notation on a number line
- Equivalence: $0.3 = 0.30 = 3/10 = 30/100$
- Comparing decimals to hundredths
- Adding tenths and hundredths by converting to a common denominator
- Converting within a measurement system (km to m, kg to g, hr to min)
- Solving measurement word problems including money and time
- Points, lines, rays, segments, and angles
- Measuring and drawing angles with a protractor
- Additive angle measure; finding an unknown angle
- Classifying triangles and quadrilaterals by angles and sides
- Lines of symmetry

LEARNING OBJECTIVES

- 1 Write a fraction with denominator 10 or 100 as a decimal and locate it on a number line.
- 2 Compare two decimals to hundredths and justify with a model.
- 3 Measure an angle to the nearest degree with a protractor.
- 4 Find an unknown angle in a decomposed figure by addition or subtraction.
- 5 Classify a triangle by both its sides and its angles.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Converts between fraction and decimal forms for tenths and hundredths in 9 of 10 trials.
- Measures angles within ± 2 degrees in 4 of 5 trials.
- Classifies triangles and quadrilaterals correctly in 8 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
0.25 is judged greater than 0.3 because $25 > 3$.	Whole-number thinking imported into decimals. Use a hundredths grid: 0.3 is 30 squares, 0.25 is 25. Also write 0.30 explicitly.
The wrong protractor scale is read, giving 130° instead of 50°.	Estimate first — acute or obtuse? — then measure. The estimate catches the scale error every time.

TEACHING NOTE

Decimals are introduced here as a notation for fractions with denominators 10 and 100, not as a separate number system. Grade 5 builds the operations; this year builds the meaning.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Weighted toward multiplication fluency and fraction meaning; identifies gaps that must be closed before Unit 2.
Unit Check	End of each unit · 25 min	Unit mastery criteria plus one extended reasoning item.
Fluency Probe	Every 2nd lesson · 3 min	Multiplication and division facts; continues all year.
Fraction Diagnostic	Before Unit 4 and after Unit 5	Equivalence, comparison, and like-denominator operations.
Mid-Year Assessment	After Unit 3	Cumulative Units 1-3; heavily weighted toward multi-digit multiplication and division.
End-of-Course Assessment	After Unit 6	All course outcomes; determines Grade 5 placement.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 50-60 min	Full course in approximately 45-50 weeks.
Accelerated	2 lessons/week	Full course in approximately 24-28 weeks.
Intensive / Catch-Up	2-3 lessons/week	18-22 weeks with a diagnostic-driven unit order.
Summer Program	3 lessons/week, 6 weeks	Multi-digit operations intensive (Units 2-3) or fraction intensive (Units 4-5).

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Twenty-five minutes per lesson: five to seven problems, of which at least two are word problems and one is cumulative review. Fact fluency practice continues in short daily doses. Students are asked to show a model or a written strategy for at least one problem per assignment — an answer alone earns no credit on those items, because the model is what the teacher needs in order to diagnose.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- ◆ Understands that each place is ten times the place to its right.
- ◆ Reads, writes, compares, and rounds numbers to one million.
- ◆ Multiplies four-digit by one-digit and two-digit by two-digit numbers.
- ◆ Divides four-digit dividends by one-digit divisors and interprets remainders in context.
- ◆ Finds all factor pairs and identifies primes and composites.
- ◆ Explains and generates equivalent fractions with justification.
- ◆ Adds and subtracts fractions and mixed numbers with like denominators.
- ◆ Multiplies a fraction by a whole number.
- ◆ Reads, writes, and compares decimals to hundredths.
- ◆ Measures and draws angles and classifies 2-D figures by properties.
- ◆ Completed Grade 4 Mathematics — ready for Grade 5.

MATERIALS AND RESOURCES

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- Base-ten blocks extended to thousands; place-value charts to millions.
- Fraction strips, fraction circles, and a hundredths grid.
- Protractor and straightedge (physical, at home).
- Grid paper for area models — used in nearly every lesson of Units 2 and 3.
- Վարժարան Grade 4 problem bank — approximately 1,100 items with model-required tagging.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.